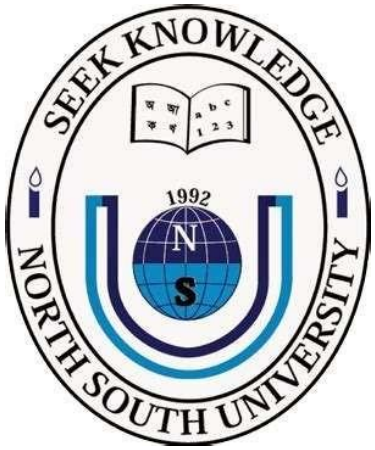
Analog Electronics II: Op-amp & 555 Timer

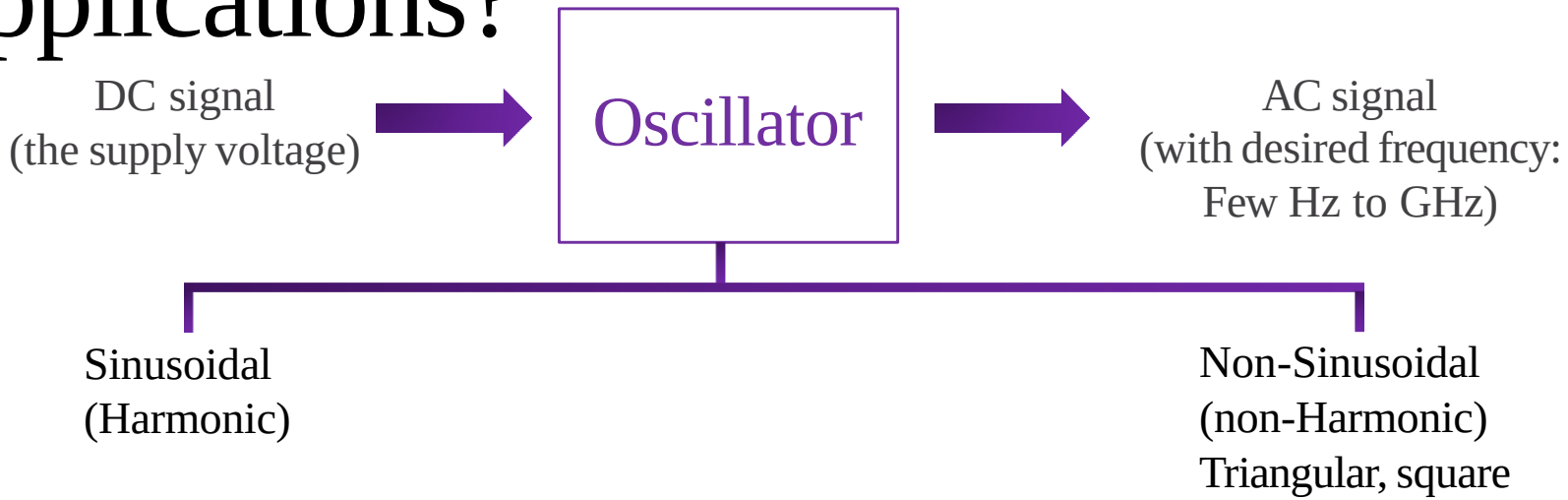


Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 13

Oscillator – What is it? Applications?



Applications:

Oscillators are used for generating:

- Clock signals
- Carrier frequency
- In signal generator

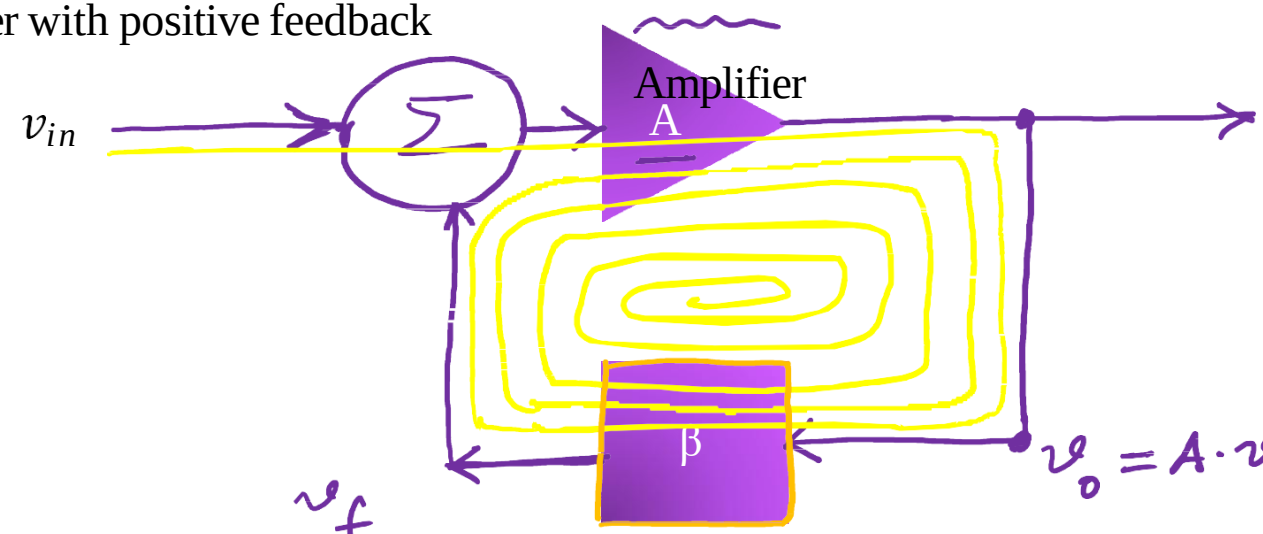
Oscillator – block diagram

$A = \frac{v_o}{v_{in}}$

A = gain

A > 1

Amplifier with positive feedback



$\beta = \frac{v_f}{v_o}$

Positive feedback path /
Regenerative feedback path /
Frequency selector Circuit /
Resonant Circuit

$v_f = \beta \cdot v_o$

$= \beta (A v_{in})$

$v_f = (A\beta) \cdot v_{in}$

$A\beta = \frac{v_f}{v_{in}}$

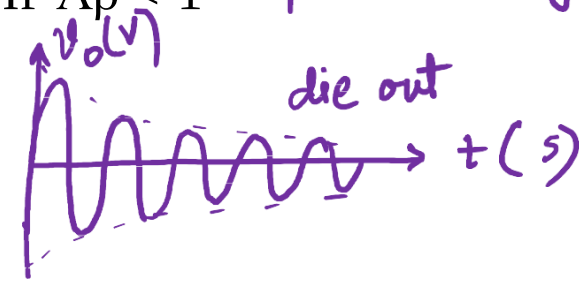
loop gain

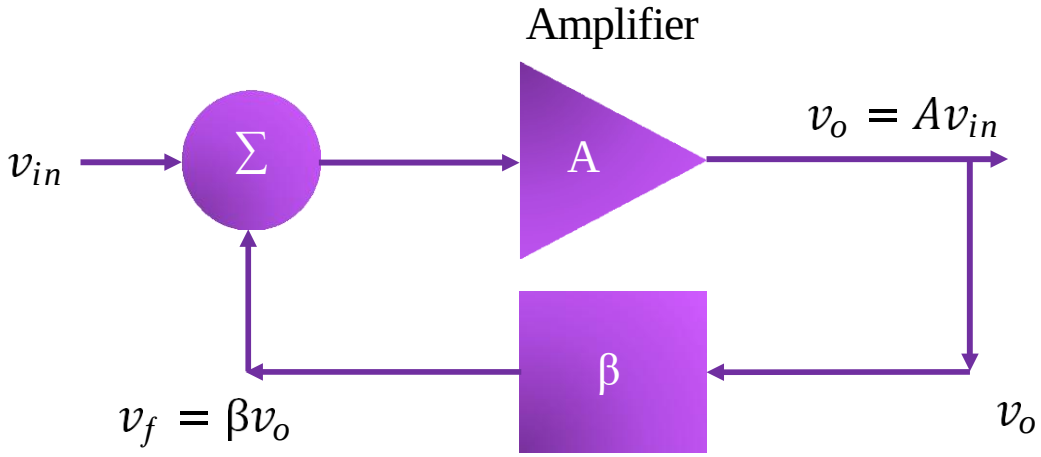
β = feedback fraction

β < 1

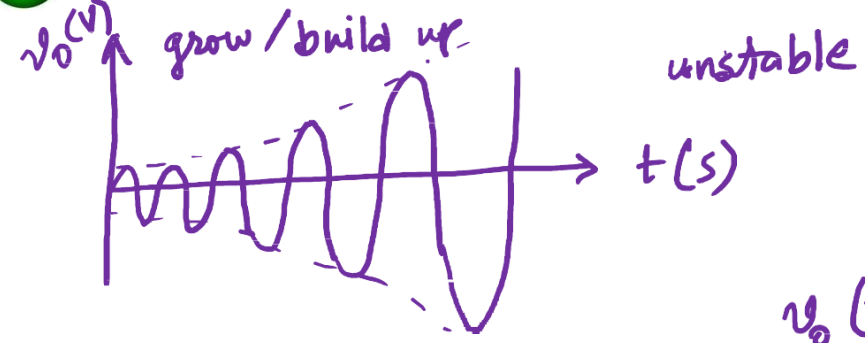
Loop gain

- If $A\beta < 1$ — β dominating

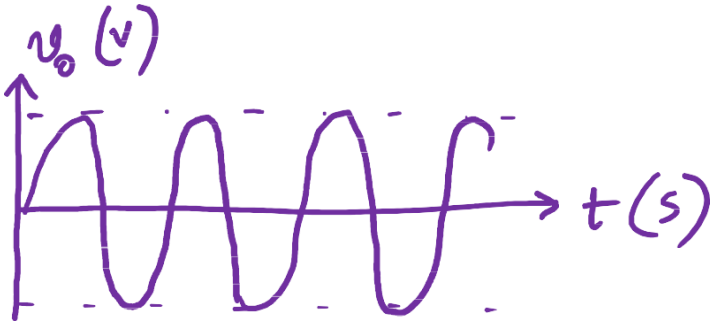




- If $A\beta > 1$ — A dominating

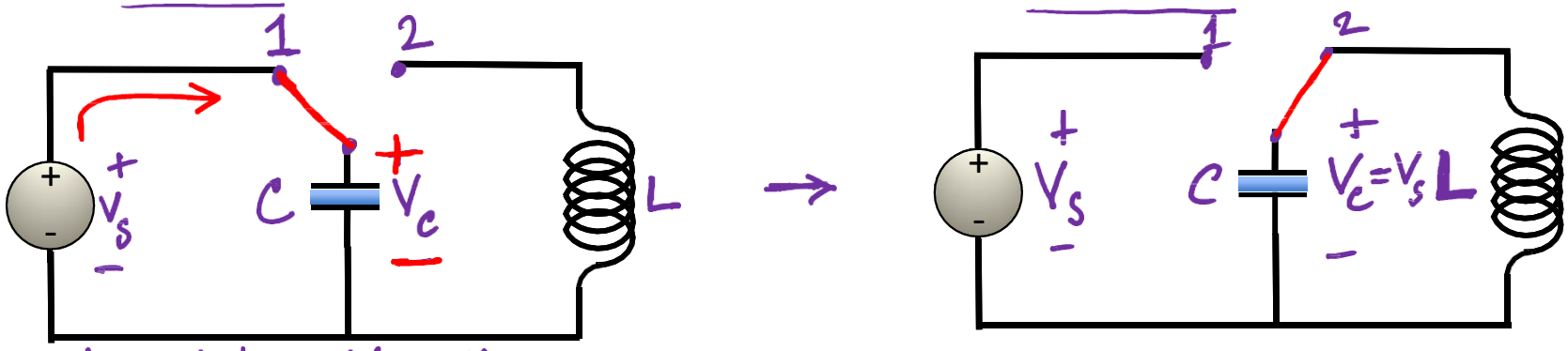


- If $A\beta = 1$ (for sustained osc^n)

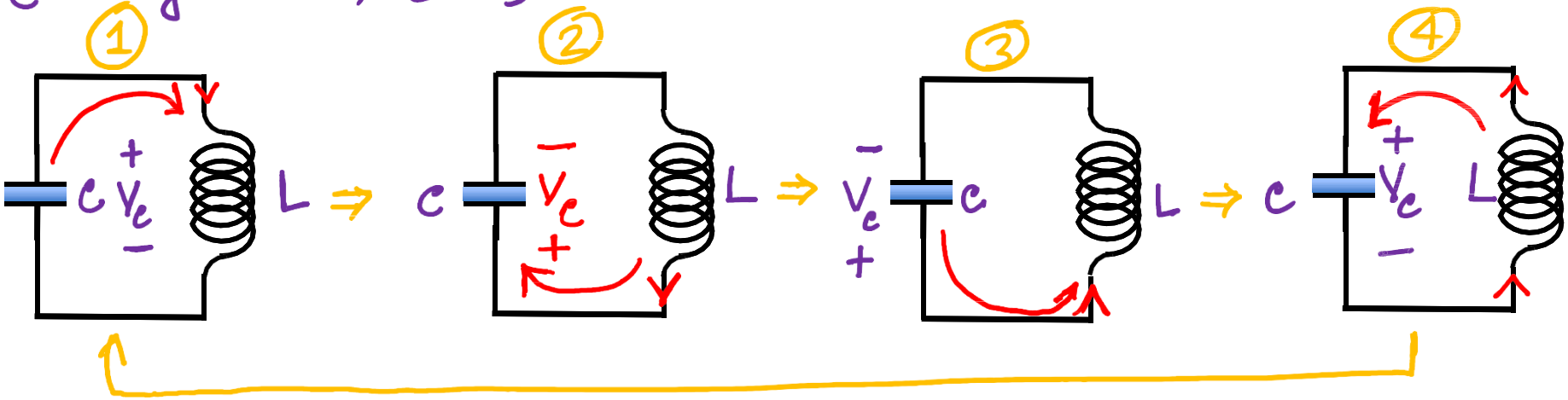


LC tank circuit

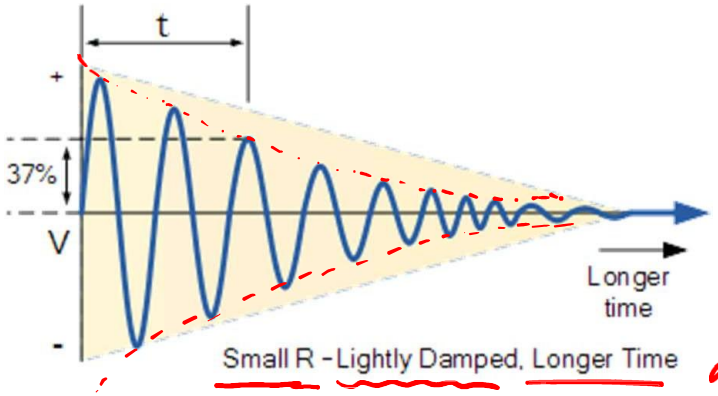
- ✓ The capacitor stores energy in the form of an electrostatic field and which produces a potential (*static voltage*) across its plates.
- ✓ The inductive coil stores its energy in the form of an electromagnetic field.

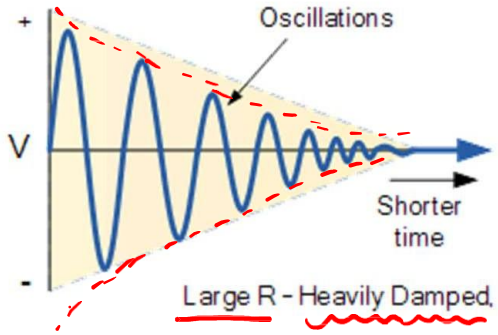


@ steady state, $V_c = V_s$



LC tank resonance





$X_L = 2\pi f L$

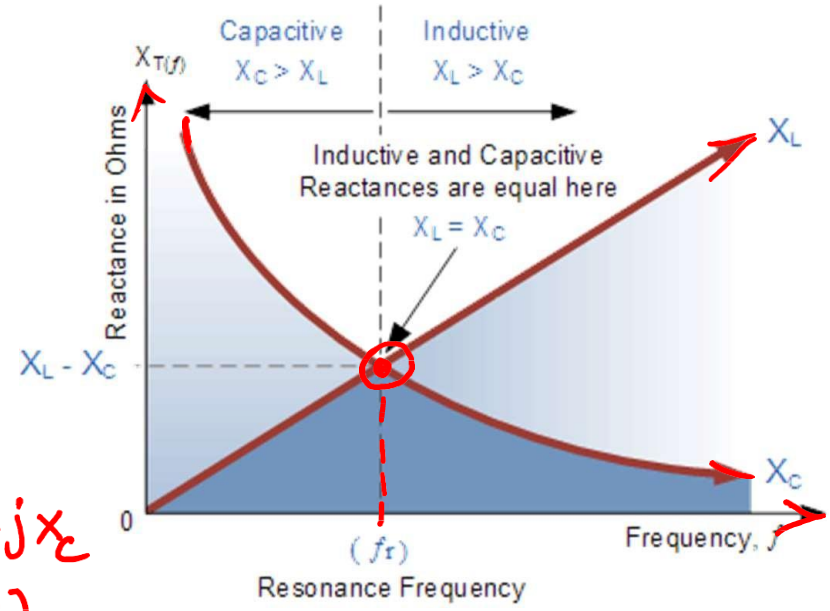
$X_C = \frac{1}{2\pi f C}$

$Z = R + jX_L - jX_C$

$Z = R$ (real)

v_o & v_{in} in phase

@ $f = f_r$



@ $f = f_r$

$X_L = X_C$

$\Rightarrow 2\pi f_r L = \frac{1}{2\pi f_r C}$

$\Rightarrow f_r^2 = \frac{1}{(2\pi)^2 LC}$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$



Further reading:

ELECTRONICS HUB

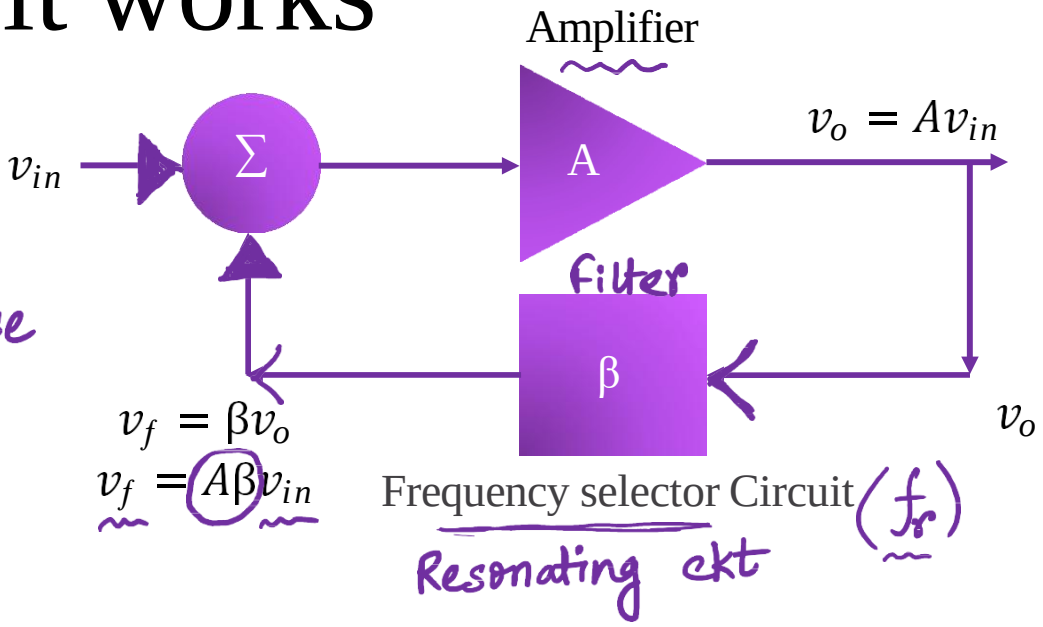
ElectronicsTutorials

Oscillator – how it works

- If the phase shift introduced by the amplifier and feedback circuit is 0 deg,

v_f & v_{in} ⇒ in the same phase

$\angle A\beta = 0^\circ$



- After the first cycle, if v_{in} is removed,

v_f works as the i/p

- Whether we will have sustained oscillations or not,

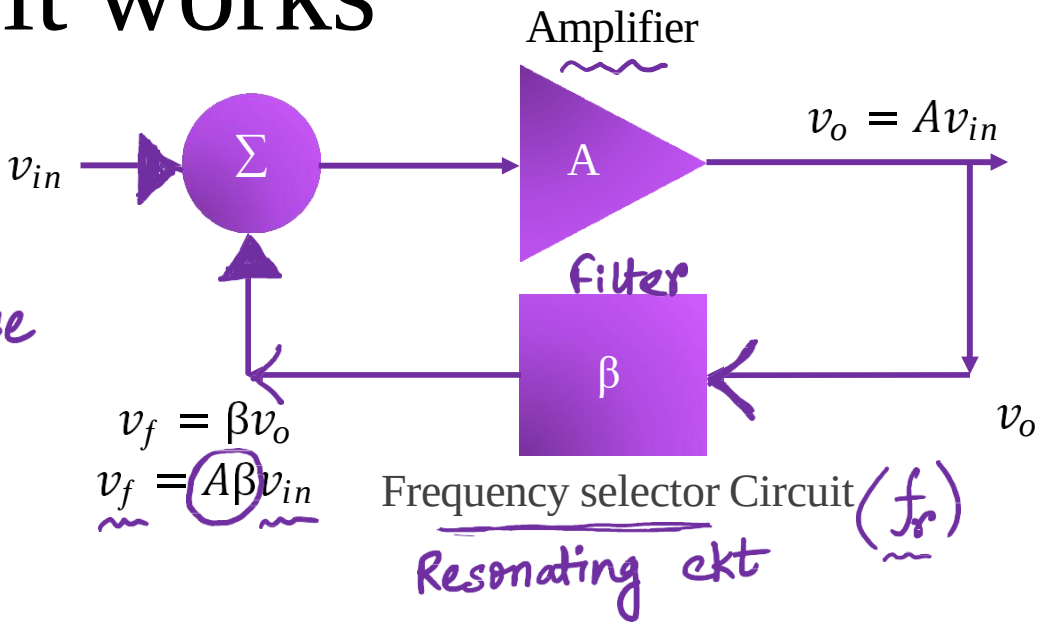
$A\beta \begin{cases} > 1 \\ < 1 \\ = 1 \end{cases}$

Oscillator – how it works

- If the phase shift introduced by the amplifier and feedback circuit is 0 deg,

v_f & v_{in} ⇒ in the same phase

$\angle A\beta = 0^\circ$



- After the first cycle, if v_{in} is removed,

v_f works as the i/p

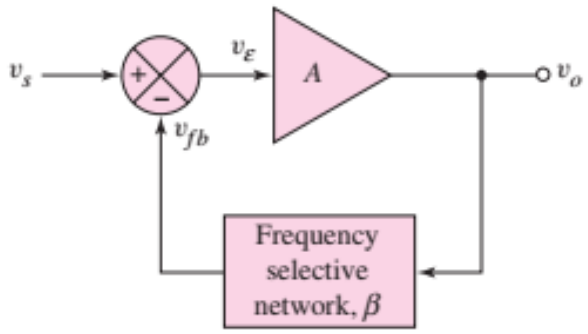
- Whether we will have sustained oscillations or not,

$A\beta \begin{cases} > 1 \\ < 1 \\ = 1 \end{cases}$

Sinusoidal Oscilator

Chapter 15.2 from Microelectronic Circuit Analysis and Design by Donald A Neaman 4th Ed.

The basic oscillator consists of an amplifier and a frequency-selective network connected in a feedback loop.



Block diagram of the fundamental feedback circuit

For the circuit shown, the ideal closed-loop transfer function is given by

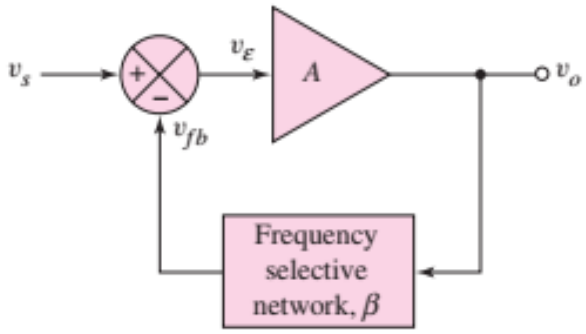
$$A_f(s) = \frac{A(s)}{1 + A(s)\beta(s)}$$

The feedback transfer function β is the principal portion of the loop gain that is dependent on frequency.

Oscillators Basics

The loop gain of the feedback circui $T(s) = A(s)\beta(s)$

If T(s) = −1, the closed-loop transfer function goes to infinity, which means that the circuit can have a finite output for a zero input signal.



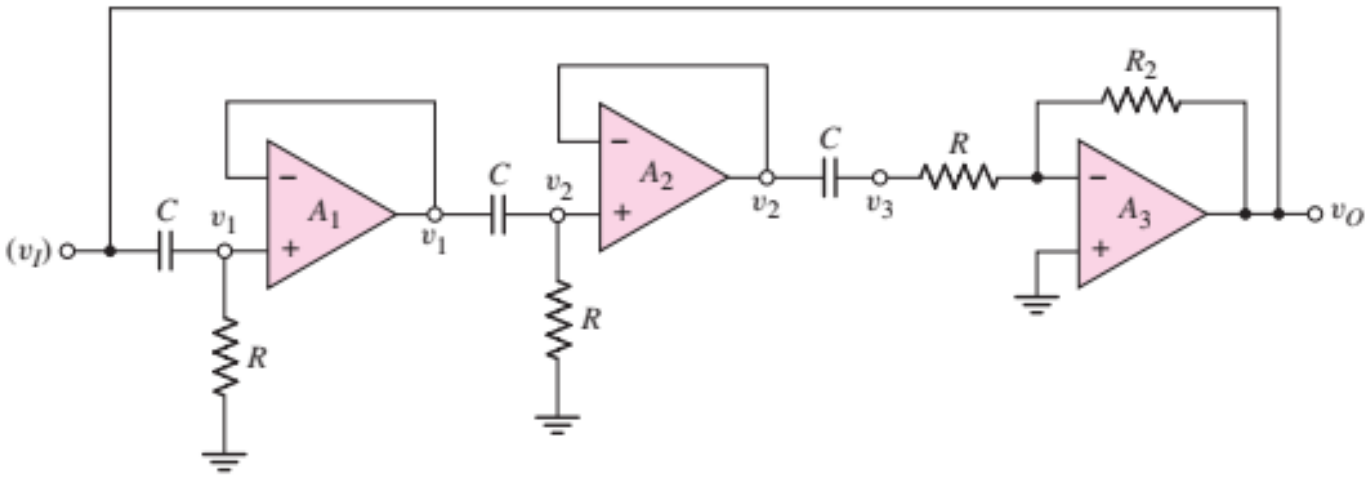
Assume that T(s) ≈ −1 so that positive feedback exists over a particular frequency range.

The condition for oscillation is that, at a specific frequency, we have

$$T(j\omega_o) = A(j\omega_o)\beta(j\omega_o) = -1$$

The condition that $T(j\omega_o) = -1$ is called the **Barkhausen criterion.**

Phase-Shift Oscillator

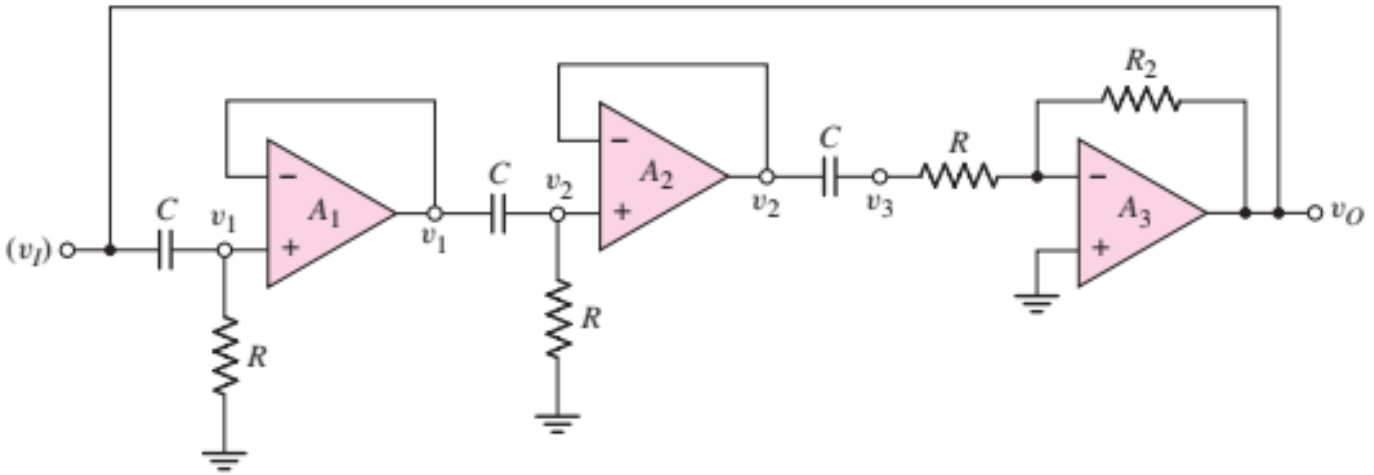


The basic amplifier of the circuit is the op-amp A3, which is connected as an inverting amplifier with its output connected to a three-stage RC filter. The voltage followers in the circuit eliminate loading effects between each RC filter stage.

The transfer function of the first RC network is

$$v_1 = \left(\frac{sRC}{1+sRC}\right)(v_I)$$

Phase-Shift Oscillator



$$\frac{v_3}{(v_I)} = \left(\frac{sRC}{1+sRC}\right)^3 = \beta(s)$$ where $\beta(s)$ is the feedback transfer function.

The amplifier gain $A(s)$ $$A(s) = \left|\frac{v_O}{v_3}\right| = \frac{R_2}{R}$$

The loop gain is then

$$T(s) = A(s)\beta(s) = \left(\frac{R_2}{R}\right)\left(\frac{sRC}{1+sRC}\right)^3$$

$$T(s) = A(s)\beta(s) = \left(\frac{R_2}{R}\right)\left(\frac{sRC}{1+sRC}\right)^3$$

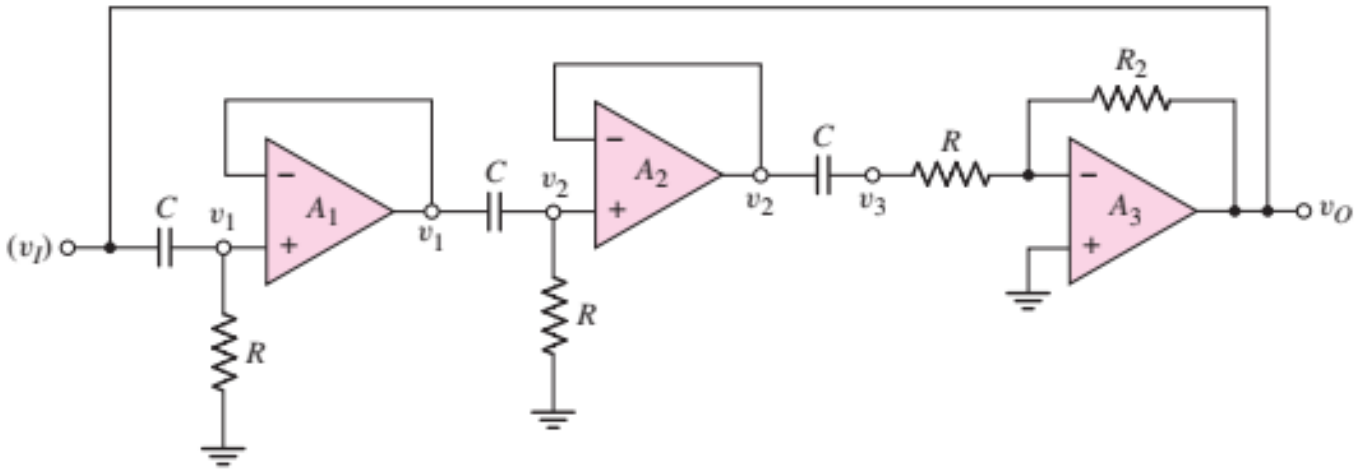
$$T(j\omega) = \left(\frac{R_2}{R}\right)\frac{(j\omega RC)^3}{(1+j\omega RC)^3}$$

$$= -\left(\frac{R_2}{R}\right)\frac{(j\omega RC)(\omega RC)^2}{[1-3\omega^2R^2C^2]+j\omega RC[3-\omega^2R^2C^2]}$$

The condition for oscillation is that |T(jωo)| = 1 and the phase of T(jωo) must be 180 degrees.

To satisfy the condition T(jωo) = −1, the imaginary component of the Equation must equal zero. Since the numerator is purely imaginary, the denominator must become purely imaginary,

$$\left[1-3\omega_o^2R^2C^2\right] = 0 \qquad \omega_o = \frac{1}{\sqrt{3}RC}$$



$$\omega_o = \frac{1}{\sqrt{3}RC}$$

where ω_o is the oscillation frequency. At this frequency,

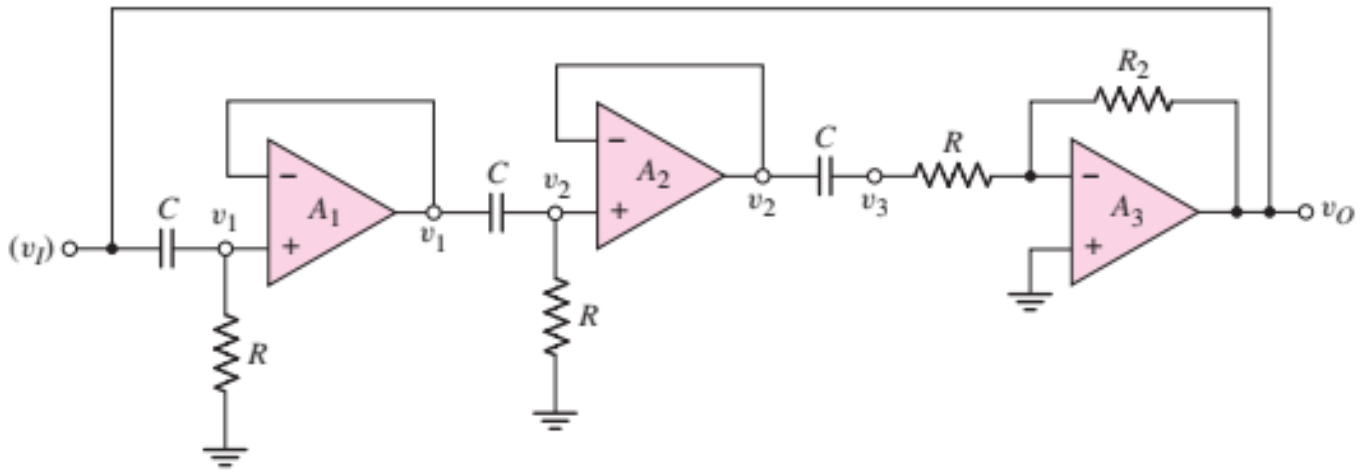
$$T(j\omega_o) = -\left(\frac{R_2}{R}\right)\frac{(j/\sqrt{3})(1/3)}{0 + (j/\sqrt{3})[3 - (1/3)]} = -\left(\frac{R_2}{R}\right)\left(\frac{1}{8}\right)$$

the condition $T(j\omega_o) = -1$ is satisfied when $\dfrac{R_2}{R} = 8$

Example Problem 15.4

Determine the oscillation frequency and required amplifier gain for a phase-shift oscillator.

Consider the phase-shift oscillator in the Figure with parameters C = 0.1 μF and R = 1 kΩ.



Solution:

The oscillation frequency is $$f_o = \frac{1}{2\pi\sqrt{3}RC} = \frac{1}{2\pi\sqrt{3}(10^3)(0.1 \times 10^{-6})} = 919\,\text{Hz}$$

The minimum amplifier gain magnitude is 8; therefore, the minimum value of R_2 is 8 kΩ

Practice Problem 15.24

Consider the phase shift oscillator in Figure 15.15. (a) For $R = 20\,\text{k}\Omega$ and $C = 0.001\,\mu\text{F}$, determine the frequency of oscillation. What is the required value of R_2? (b) Design the circuit such that the frequency of oscillation is $f_o = 25\,\text{kHz}$. Let $R = 20\,\text{k}\Omega$.

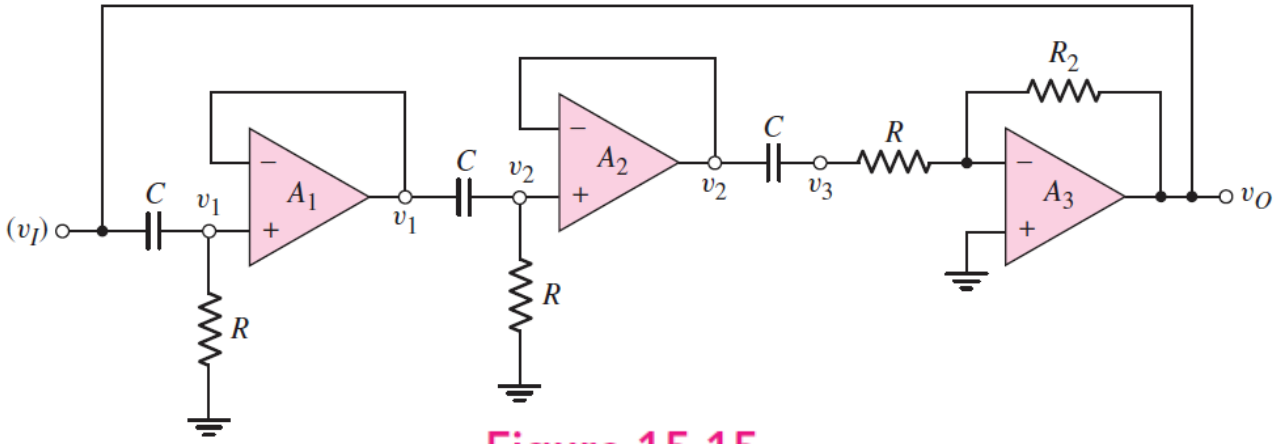


Figure 15.15

Solution:

$$\text{(a)}\quad f_o = \frac{1}{2\pi\sqrt{3}RC} = \frac{1}{2\pi\sqrt{3}\left(20\times10^{3}\right)\left(0.001\times10^{-6}\right)}$$

$$f_o = 4.59\,\text{kHz},\quad R_2 = 8R = 160\,\text{k}\Omega$$

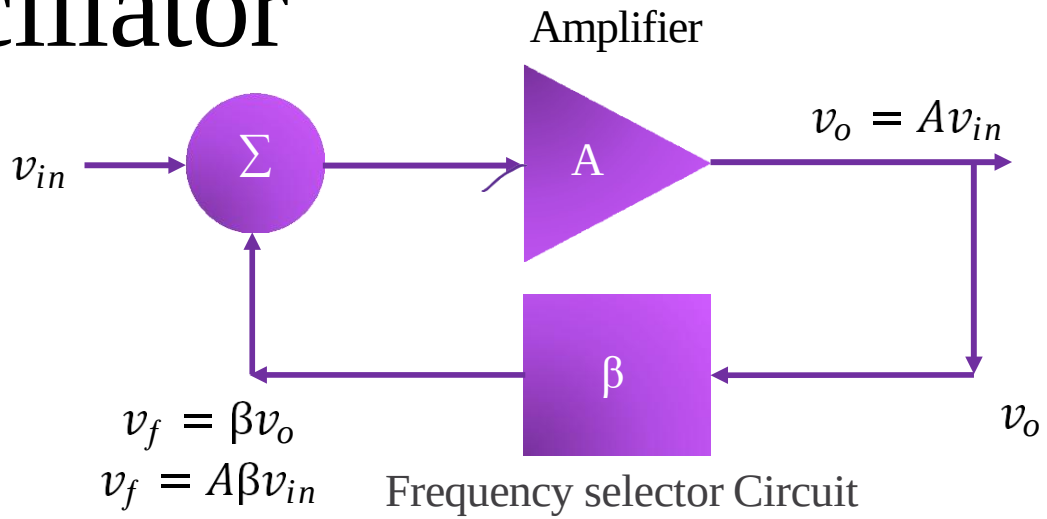
$$\text{(b)}\quad C = \frac{1}{2\pi\sqrt{3}Rf_o} = \frac{1}{2\pi\sqrt{3}\left(20\times10^{3}\right)\left(25\times10^{3}\right)}$$

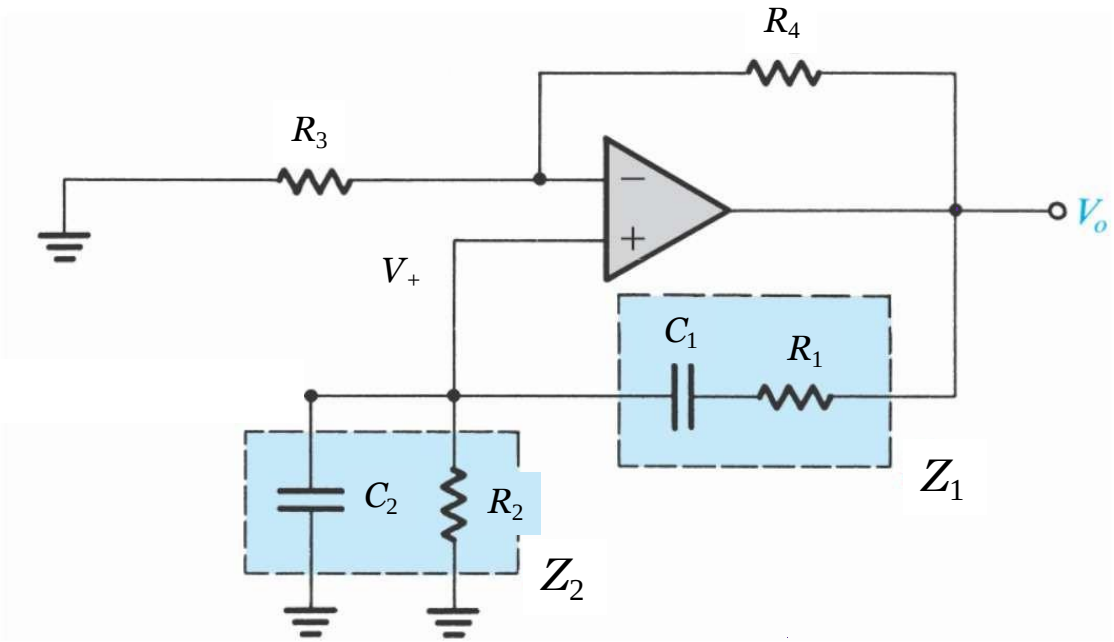
$$C = 184\,\text{pF},\quad R_2 = 160\,\text{k}\Omega$$

Wein Bridge Oscillator

- ✓ Harmonic oscillator

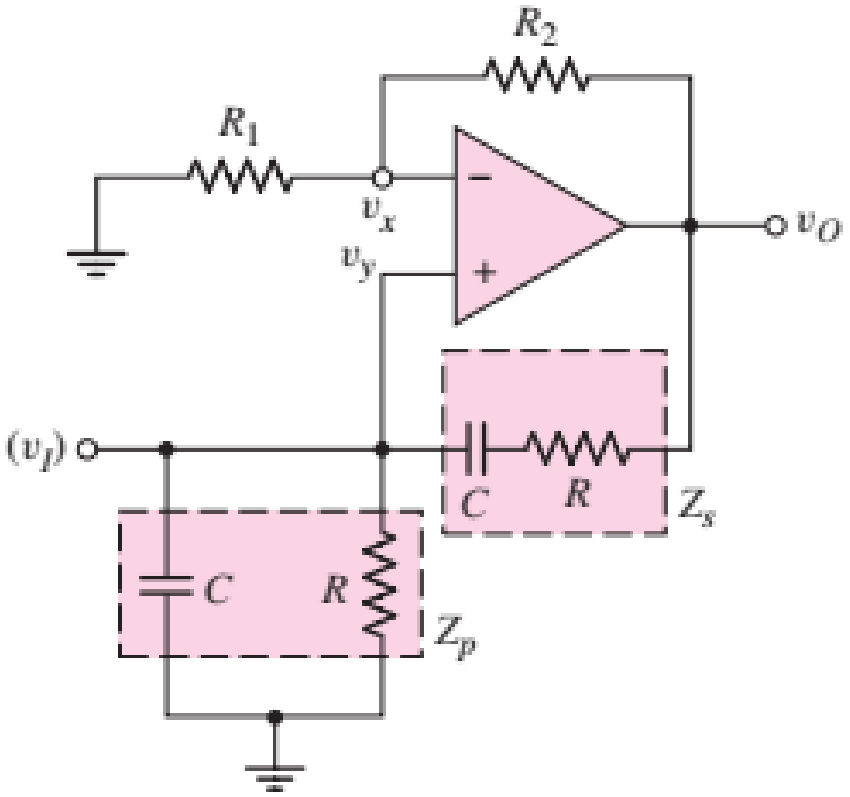
  (generates sinusoidal waveform)
- ✓ Feedback network: RC
- ✓ In audio frequency range





Wien-Bridge Oscillator

The circuit consists of an op-amp connected in a noninverting configuration and two RC networks connected as the frequency-selecting feedback circuit.

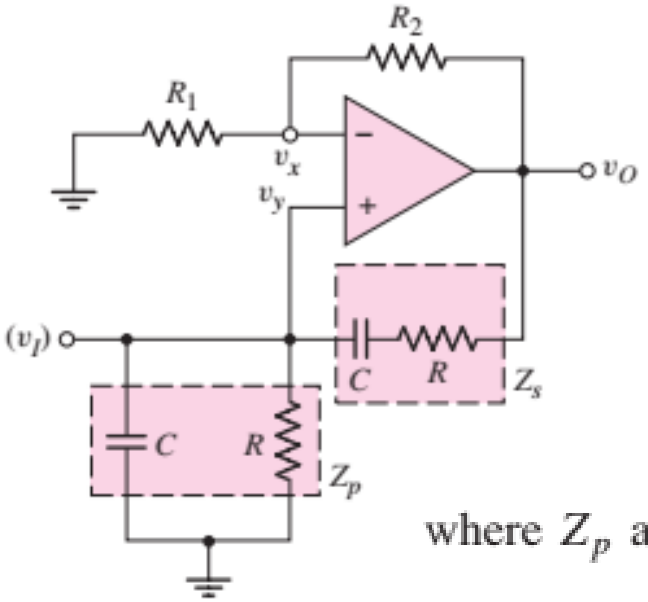


Since the noninverting amplifier introduces zero phase shift, the frequency-selective feedback circuit must also introduce zero phase shift to create the positive feedback condition.

The loop gain is the product of the amplifier gain and the feedback transfer function,

$$T(s) = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{Z_p}{Z_p + Z_s}\right)$$

Wien-Bridge Oscillator



$$T(s) = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{Z_p}{Z_p + Z_s}\right)$$

where Z_p and Z_s are the parallel and series RC network impedances.

$$Z_p = \frac{R}{1 + sRC} \qquad Z_s = \frac{1 + sRC}{sC}$$

Combining Equations $T(s) = \left(1 + \frac{R_2}{R_1}\right)\left[\frac{1}{3 + sRC + (1/sRC)}\right]$

Since this circuit has no explicit negative feedback

the condition for oscillation is $T(j\omega_o) = 1$

Calculation of *T(s)= A(s)* β*(s)*

$$\beta(s) = \left(\frac{Z_p}{Z_p + Z_s}\right) = \frac{\frac{R}{1+sRC}}{\frac{sRC+1}{sC} + \frac{R}{1+sRC}}$$

$$\beta(s) = \frac{\frac{R}{1+sRC} \cdot (1+sRC) \cdot sC}{\frac{sRC+1}{sC} \cdot (1+sRC) \cdot sC \; + \; \frac{R}{1+sRC} \cdot (1+sRC) \cdot sC}$$

$$\beta(s) = \frac{RsC}{(sRC+1)(1+sRC) \quad + \quad RsC} = \frac{RsC}{(sRC+1)^2 + RsC}$$

$$= \frac{RsC}{s^2R^2C^2 + 2sRC + 1 \; + \; RsC}$$

$$= \frac{sRC}{s^2R^2C^2 + 3sRC + 1} = \frac{1}{sRC + 3 + \frac{1}{sRC}}$$

the condition for oscillation is

$$T(j\omega_o) = 1 = \left(1 + \frac{R_2}{R_1}\right)\left[\frac{1}{3 + j\omega_o RC + (1/j\omega_o RC)}\right]$$

Since T(jωo) must be real, the imaginary component of the above Equation must be zero;

$$j\omega_o RC + \frac{1}{j\omega_o RC} = 0 \qquad \omega_o = \frac{1}{RC}$$

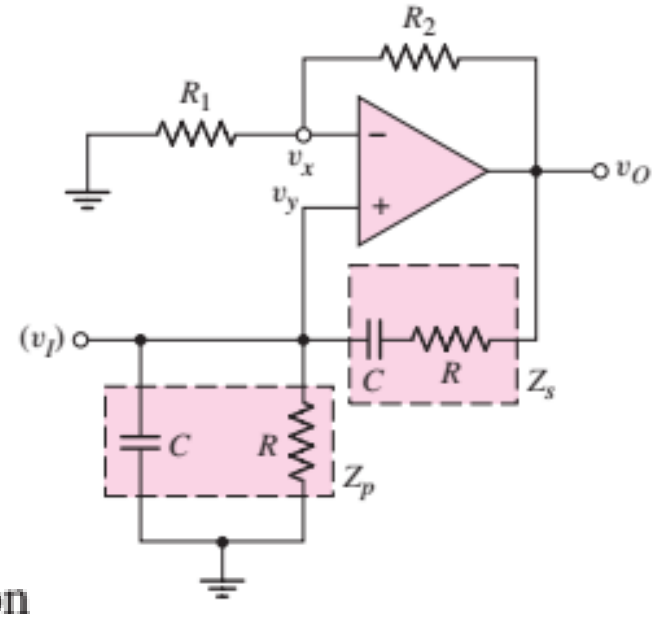
The magnitude condition is then

$$1 = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{1}{3}\right) \qquad \frac{R_2}{R_1} = 2$$

Example Problem 15.5

Design a Wien-bridge circuit to oscillate at a specified frequency.

Specifications: Design the Wien-bridge oscillator shown in the Figure to oscillate at f_o = 20 kHz.

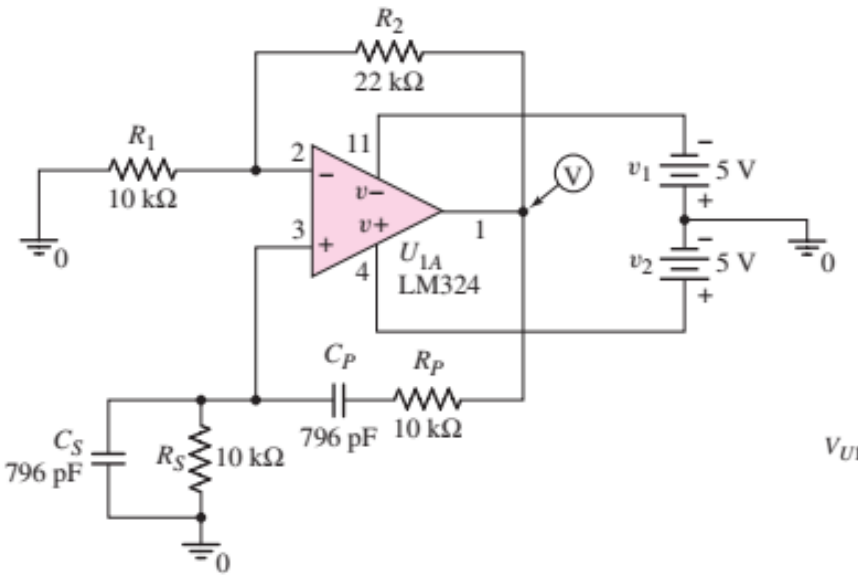


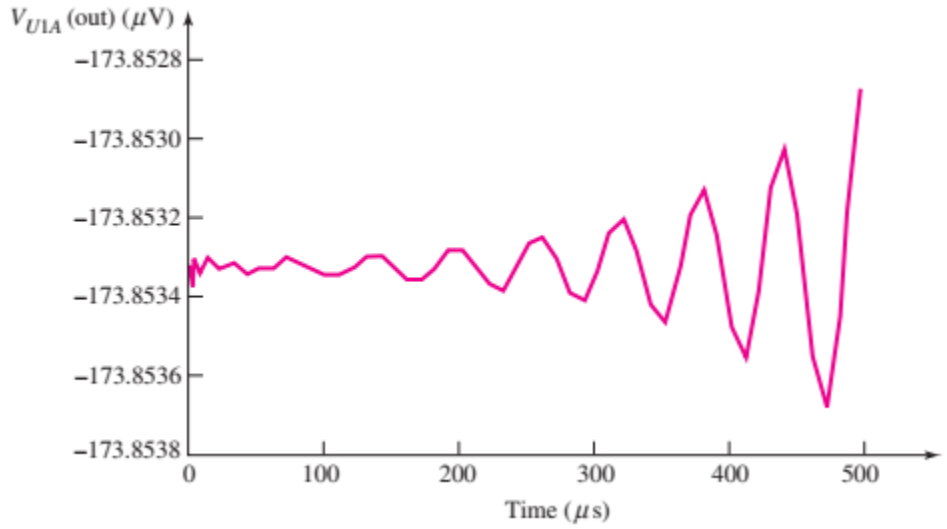
Solution: The oscillation frequency given by Equation

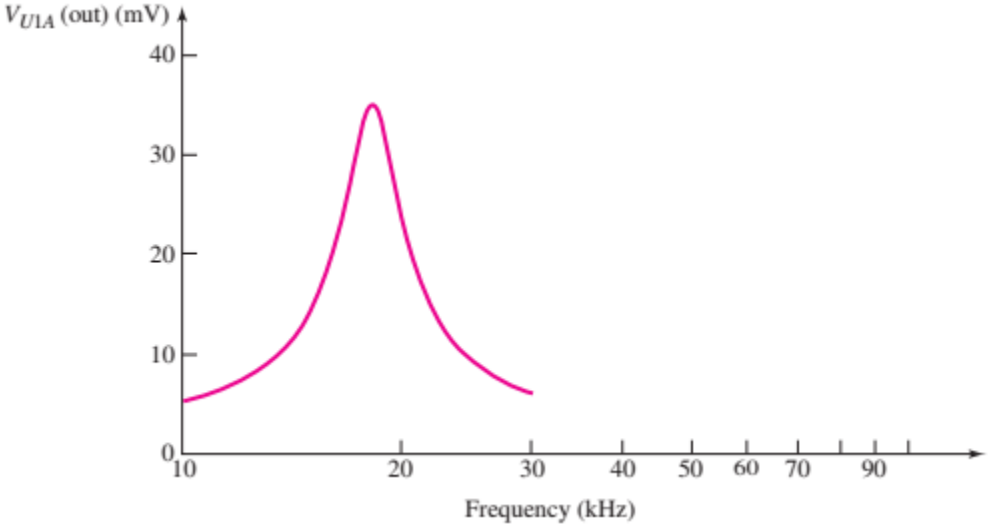
$$RC = \frac{1}{2\pi f_o} = \frac{1}{2\pi(20 \times 10^3)} = 7.96 \times 10^{-6}$$

A 10 kΩ resistor and 796 pF capacitor satisfy this requirement. Since the amplifier resistor ratio must be $R_2/R_1 = 2$, we could, for example, have $R_2 = 20\,\text{k}\Omega$ and $R_1 = 10\,\text{k}\Omega$, which would satisfy the requirement.

Simulation of Problem 15.5



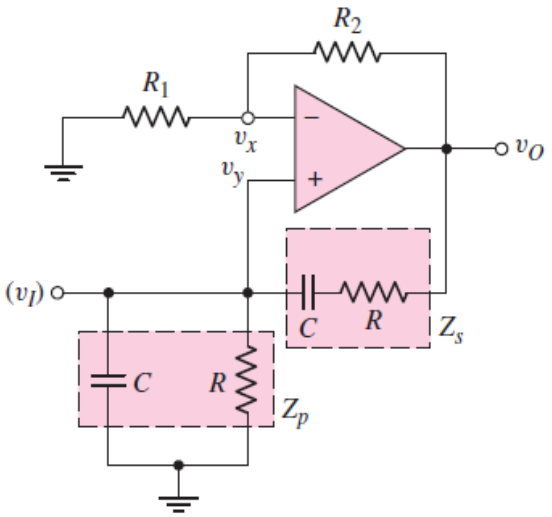




Practice Problem 15.34

Design the Wein-bridge oscillator to oscillate at $f_o = 35\,\text{kHz}$. Choose appropriate component values.

Solution:



$$f_o = \frac{1}{2\pi RC}$$

$$RC = \frac{1}{2\pi f_o} = \frac{1}{2\pi\left(35\times10^3\right)} = 4.547\times10^{-6}$$

Let $C = 0.001\,\mu\text{F}$, then $R = 4.55\,\text{k}\Omega$

Set $\dfrac{R_2}{R_1} = 2$